

EX No:1	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK
DATE:	

Aim:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

Algorithm:

Algorithm 1: Installing Python

1. Download Python from python.org
2. Run the installer
3. Select "Add Python to PATH"
4. Click Install Now
5. Verify installation using: `python --version`
6. Observe installed Python version

Algorithm 2: Installing Jupyter Notebook

1. Open Command Prompt
2. Install Jupyter `pip install notebook`
3. Launch Jupyter Notebook `jupyter notebook`
4. Browser automatically opens Notebook Dashboard.

Algorithm 3: Creating a Notebook

1. Open Jupyter Dashboard
2. Select New → Python 3
3. Create Notebook
4. Rename Notebook
5. Create Code and Markdown Cells
6. Execute cells using Shift + Enter

Program:

Program 1 — Simple Python Program

```
>>> print("Data Analytics Lab")
```

Output:

```
Data Analytics Lab
```

Program 2 — Arithmetic Operations

```
>>> a=20
>>> b=10
>>> print("Addition:",a+b)
>>> print("Subtraction:",a-b)
>>> print("Multiplication:",a*b)
>>> print("Division:",a/b)
```

Output:

```
Addition: 30
```

```
Subtraction: 10
```

```
Multiplication: 200
```

```
Division: 2.0
```

Program 3 — Display Student Details

```
>>> name="John"
>>> dept="AIML"
>>> cgpa=8.9
>>> print("NAME:",name)
>>> print("DEPARTMENT:",dept)
>>> print("CGPA:",cgpa)
```

Output:

```
NAME: John
```

```
DEPARTMENT: AIML
```

```
CGPA: 8.9
```

Program 4 — Creating Markdown Cell

```
# Data Analytics Lab
## Experiment 1
### Jupyter Notebook
```

Then press **Shift + Enter**.

Data Analytics Lab

Experiment 1

Jupyter Notebook

Program 5 — Using NumPy Library

```
import numpy as np
a = np.array([10,20,30,40,50])
print(a)
print("Mean =",np.mean(a))
print("Sum =",np.sum(a))
```

Output:

```
[10 20 30 40 50]
Mean = 30.0
Sum = 150
```

Program 6 — Using Pandas Library

```
import pandas as pd
data = {
    'Name': ['John', 'David', 'Alex'], 'Marks': [85,90,95]
}
df = pd.DataFrame(data)
print(df)
```

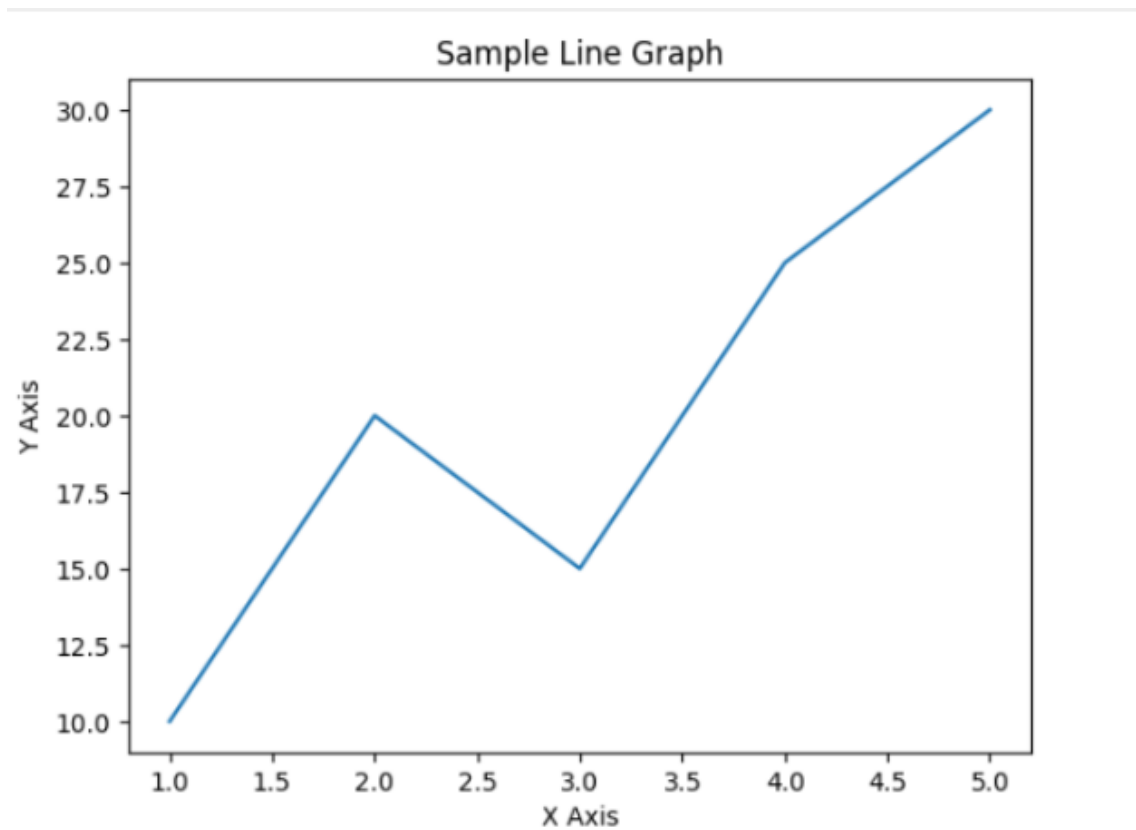
Output:

	Name	Marks
0	John	85
1	David	90
2	Alex	95

Program 7 — Data Visualization

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]
plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

Output:



Result: